

Series and infinite products – 5

Double and multiple series

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Theoretical minimum.

Theorem 1 (Tonelli and Fubini for series). For $a_{mn} \geq 0$ the two iterated sums and the supremum over finite subsets of $\mathbb{N}^2$ are always equal, finite or not. For general terms, if one of the iterated sums of $|a_{mn}|$ is finite, then all three coincide. Without this the order matters: for

$$a_{mn} = \frac{m - n}{(m + n)^3}$$

the two iterated sums are non-zero and opposite.

Theorem 2 (Cauchy product and Mertens). If $\sum a_n$ and $\sum b_n$ converge and at least one of them absolutely, then $\sum c_n$ with $c_n = \sum_{k=0}^n a_k b_{n-k}$ converges to the product of the sums. For two conditionally convergent series the product may diverge, as for $a_n = b_n = \frac{(-1)^n}{\sqrt{n+1}}$.

Theorem 3 (summation along level sets). Rearranging $\mathbb{N}^2$ by the value of a function of (m, n) turns a double sum into a single one with an arithmetic weight:

$$\sum_{m,n \geq 1} f(mn) = \sum_{k \geq 1} d(k) f(k), \quad \sum_{m,n \geq 1} f(m+n) = \sum_{k \geq 2} (k-1) f(k),$$

and for non-negative terms no justification beyond Theorem 1 is needed.

Fact 1 (how a k -fold sum collapses). A factor depending on $n_1 + \dots + n_k$ is made separable by an integral: $\frac{1}{n_1 + \dots + n_k + 1} = \int_0^1 x^{n_1 + \dots + n_k} dx$. After the interchange the integrand is a product of k equal single sums — this one line solves most multiple-sum problems.

Fact 2 (multiple zeta values). For $s_1 \geq 2$ put $\zeta(s_1, s_2) = \sum_{m > n \geq 1} m^{-s_1} n^{-s_2}$.

Splitting $\mathbb{N}^2$ into $m > n$, $m < n$ and $m = n$ gives the shuffle relation $\zeta(a)\zeta(b) = \zeta(a, b) + \zeta(b, a) + \zeta(a + b)$, and Euler's theorem says $\zeta(2, 1) = \zeta(3)$.

1. Prove Theorem 1 in both parts and verify that for $a_{mn} = \frac{m-n}{(m+n)^3}$ the two iterated sums are different; then give an example where one iterated sum exists and the other does not.
2. (IMC 2016) Let $(x_1, x_2, \dots)$ be a sequence of positive real numbers satisfying $\sum_{n=1}^{\infty} \frac{x_n}{2^{n-1}} = 1$. Prove that $\sum_{k=1}^{\infty} \sum_{n=1}^k \frac{x_n}{k^2} \leq 2$.
3. (VJIMC 2011) Let k be a positive integer. Compute

$$\sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \cdots \sum_{n_k=1}^{\infty} \frac{1}{n_1 n_2 \cdots n_k (n_1 + \cdots + n_k + 1)}.$$

4. (VJIMC 2016) Find the value of the sum $\sum_{n=1}^{\infty} A_n$, where

$$A_n = \sum_{k_1=1}^{\infty} \cdots \sum_{k_n=1}^{\infty} \frac{1}{k_1^2} \cdot \frac{1}{k_1^2 + k_2^2} \cdots \frac{1}{k_1^2 + \cdots + k_n^2}.$$

5. (SEEMOUS) Let $\mathcal{F}$ be the family of all non-empty finite subsets of $\mathbb{N} \cup \{0\}$. Find all real numbers a for which the series $\sum_{A \in \mathcal{F}} \frac{1}{\sum_{k \in A} a^k}$ is convergent.